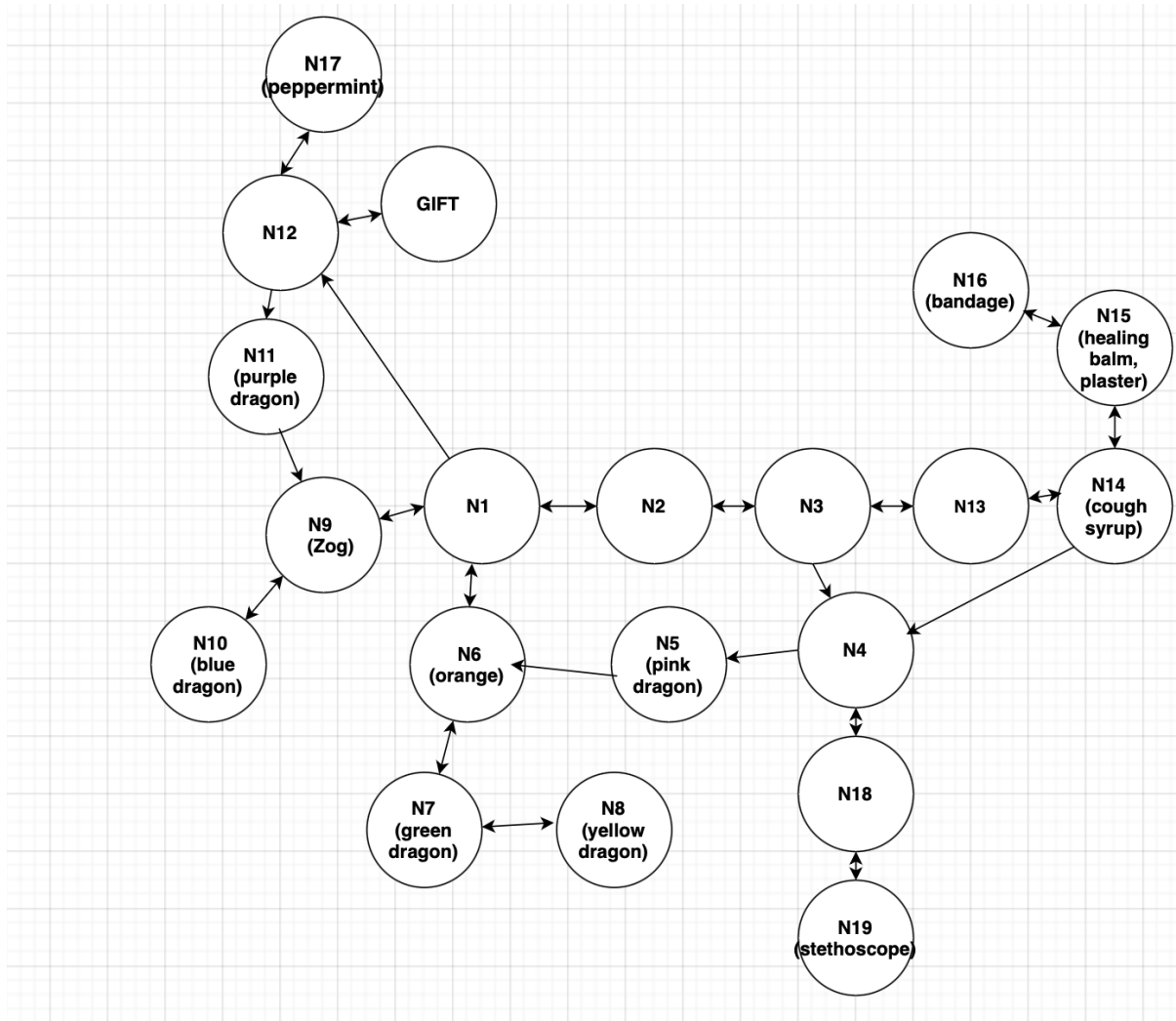


Informatics 2D Coursework Report

1. (5 marks) Task 2.1 Design



My design for harder problem. I have increased the number of nodes from 12 to 19. I have added 2 extra patients and increased a total delivery request from 6 to 9. I also put items further away from patients (nodes N16, N15, N14, N19, N17) therefore it requires longer routes for Zog to take. As a result, it will significantly increase the search space which will result planner to explore more states as requested.

2. (10 marks) Task 2.2 Evaluation

Initially my hard problem was designed with 8 patients and 24 total deliveries. But during testing, high wg, low wh failed to run within a reasonable time. Therefore, I kept scaling

down my problem, slowly bit by bit. So, I stopped on the current design with 9 total delivery requests and 6 patients as described in design.

The results with different wg and wh for my problem. (actual results showed in the end of the document)

Wg (cost)	Wh (heuristic)	Runtime (s)	Plan length
1	1	46.35(s)	85
1	5	0.00(s)	95
1	10	0.00(s)	101
0	1	0.00(s)	167
5	1	68.94(s)	85
10	1	67.31(s)	85
1	0	18.93(s)	85

So, my results show clear tradeoff between search time and quality of the plan. The higher wh (heuristics) significantly reduce run time but tend to produce longer plans. On other hand wg (cost) maintain plan length at 85 (I assume it the lowest plan length possible), but it increases runtime much more. Setting wg to 0 produces the longest plan (167 steps) despite fast search, that demonstrates that cost weight leads to poor quality solutions. The default values (wg=1, wh=5) provide a good balance between search speed and plan quality."

3. (15 marks) Task 3.4 Your Extension

In the real medical emergencies, patients have different level of urgency to get seen, conditions of the patient can get worse if the treatment is delayed. So, I decided to implement timer function, so certain patients have bigger urgency to be seen than others.

Therefore, this constraint means not all plans are valid and Zog needs to prioritise to treat some patients early.

To implement it I added numeric fluent time tracks how many moves have passed since the plan started, incrementing each time Zog moves. A second fluent max-urgency defines the maximum number of moves allowed before each patient's condition becomes critical. Delivery is only possible if $\text{time} < \text{max-urgency}$

Results from terminal for 2.2

```
./ff -E -g 1 -h 5 -o domain-1.pddl -f problem-1-hard.pddl

[barnett]s2634187: ./ff -E -g 1 -h 1 -o domain-1.pddl -f problem-1-
hard.pddl

ff: parsing domain file
domain 'ZOGS-WORLD' defined
... done.
ff: parsing problem file
problem 'ZOGS-PROBLEM' defined
... done.

no metric specified. plan length assumed.

checking for cyclic := effects --- OK.

ff: search configuration is  best-first on  $1*g(s) + 1*h(s)$  where
    metric is  plan length

advancing to distance:  33
                        32
                        31
                        30
                        29
                        28
                        27
                        26
                        25
                        24
                        23
                        22
                        21
                        20
                        19
                        18
                        17
                        16
                        15
                        14
                        13
                        12
                        11
```

10
9
8
7
6
5
4
3
2
1
0

ff: found legal plan as follows

```
step    0: MOVE N9 ZOG N1
        1: MOVE N1 ZOG N2
        2: MOVE N2 ZOG N3
        3: MOVE N3 ZOG N13
        4: MOVE N13 ZOG N14
        5: MOVE N14 ZOG N15
        6: PICK-UP PLASTER N15 ZOG
        7: MOVE N15 ZOG N16
        8: PICK-UP BANDAGE N16 ZOG
        9: MOVE N16 ZOG N15
       10: MOVE N15 ZOG N14
       11: MOVE N14 ZOG N4
       12: MOVE N4 ZOG N18
       13: MOVE N18 ZOG N19
       14: PICK-UP STETHOSCOPE N19 ZOG
       15: MOVE N19 ZOG N18
       16: MOVE N18 ZOG N4
       17: MOVE N4 ZOG N5
       18: MOVE N5 ZOG N6
       19: MOVE N6 ZOG N7
       20: DELIVERY ZOG STETHOSCOPE GREEN-DRAGON N7
       21: MOVE N7 ZOG N6
       22: DELIVERY ZOG PLASTER ORANGE-DRAGON N6
       23: MOVE N6 ZOG N1
       24: MOVE N1 ZOG N12
       25: MOVE N12 ZOG N17
       26: PICK-UP PEPPERMINT N17 ZOG
       27: MOVE N17 ZOG N12
       28: MOVE N12 ZOG N11
       29: DELIVERY ZOG BANDAGE PURPLE-DRAGON N11
       30: MOVE N11 ZOG N9
       31: MOVE N9 ZOG N1
       32: MOVE N1 ZOG N2
       33: MOVE N2 ZOG N3
       34: MOVE N3 ZOG N13
       35: MOVE N13 ZOG N14
       36: MOVE N14 ZOG N15
       37: PICK-UP PLASTER N15 ZOG
       38: MOVE N15 ZOG N16
       39: PICK-UP BANDAGE N16 ZOG
       40: MOVE N16 ZOG N15
       41: MOVE N15 ZOG N14
       42: MOVE N14 ZOG N4
       43: MOVE N4 ZOG N18
       44: MOVE N18 ZOG N19
       45: PICK-UP STETHOSCOPE N19 ZOG
       46: MOVE N19 ZOG N18
```

```

47: MOVE N18 ZOG N4
48: MOVE N4 ZOG N5
49: DELIVERY ZOG PLASTER PINK-DRAGON N5
50: DELIVERY ZOG PEPPERMINT PINK-DRAGON N5
51: MOVE N5 ZOG N6
52: MOVE N6 ZOG N7
53: MOVE N7 ZOG N8
54: DELIVERY ZOG STETHOSCOPE YELLOW-DRAGON N8
55: DELIVERY ZOG BANDAGE YELLOW-DRAGON N8
56: MOVE N8 ZOG N7
57: MOVE N7 ZOG N6
58: MOVE N6 ZOG N1
59: MOVE N1 ZOG N2
60: MOVE N2 ZOG N3
61: MOVE N3 ZOG N13
62: MOVE N13 ZOG N14
63: MOVE N14 ZOG N15
64: MOVE N15 ZOG N16
65: PICK-UP BANDAGE N16 ZOG
66: MOVE N16 ZOG N15
67: MOVE N15 ZOG N14
68: MOVE N14 ZOG N13
69: MOVE N13 ZOG N3
70: MOVE N3 ZOG N2
71: MOVE N2 ZOG N1
72: MOVE N1 ZOG N12
73: MOVE N12 ZOG N17
74: PICK-UP PEPPERMINT N17 ZOG
75: MOVE N17 ZOG N12
76: MOVE N12 ZOG N11
77: MOVE N11 ZOG N9
78: MOVE N9 ZOG N10
79: DELIVERY ZOG PEPPERMINT BLUE-DRAGON N10
80: DELIVERY ZOG BANDAGE BLUE-DRAGON N10
81: MOVE N10 ZOG N9
82: MOVE N9 ZOG N1
83: MOVE N1 ZOG N12
84: MOVE N12 ZOG GIFT-SHOP
85: BUY-GIFT ZOG GIFT-SHOP

```

```

time spent:    0.00 seconds instantiating 54 easy, 0 hard action templates
              0.00 seconds reachability analysis, yielding 42 facts and 54
actions
              0.00 seconds creating final representation with 42 relevant
facts, 0 relevant fluents
              0.00 seconds computing LNF
              0.00 seconds building connectivity graph
              46.35 seconds searching, evaluating 415350 states, to a max
depth of 0
              46.35 seconds total time

```

=====

```

[barnett]s2634187: ./ff -E -g 1 -h 5 -o domain-1.pddl -f problem-1-
hard.pddl

```

```

ff: parsing domain file
domain 'ZOGS-WORLD' defined

```

```
... done.
ff: parsing problem file
problem 'ZOGS-PROBLEM' defined
... done.
```

no metric specified. plan length assumed.

checking for cyclic := effects --- OK.

ff: search configuration is best-first on $1*g(s) + 5*h(s)$ where
metric is plan length

```
advancing to distance:  33
                        32
                        31
                        29
                        28
                        27
                        26
                        25
                        24
                        23
                        22
                        21
                        20
                        19
                        18
                        17
                        16
                        15
                        14
                        12
                        11
                        10
                         9
                         8
                         7
                         6
                         5
                         4
                         3
                         2
                         1
                         0
```

ff: found legal plan as follows

```
step    0: MOVE N9 ZOG N1
         1: MOVE N1 ZOG N12
         2: MOVE N12 ZOG N17
         3: PICK-UP PEPPERMINT N17 ZOG
         4: MOVE N17 ZOG N12
         5: MOVE N12 ZOG N11
         6: MOVE N11 ZOG N9
         7: MOVE N9 ZOG N1
         8: MOVE N1 ZOG N2
         9: MOVE N2 ZOG N3
        10: MOVE N3 ZOG N13
        11: MOVE N13 ZOG N14
        12: MOVE N14 ZOG N15
```

13: PICK-UP PLASTER N15 ZOG
14: MOVE N15 ZOG N16
15: PICK-UP BANDAGE N16 ZOG
16: MOVE N16 ZOG N15
17: MOVE N15 ZOG N14
18: MOVE N14 ZOG N4
19: MOVE N4 ZOG N18
20: MOVE N18 ZOG N19
21: PICK-UP STETHOSCOPE N19 ZOG
22: MOVE N19 ZOG N18
23: MOVE N18 ZOG N4
24: MOVE N4 ZOG N5
25: DELIVERY ZOG PEPPERMINT PINK-DRAGON N5
26: DELIVERY ZOG PLASTER PINK-DRAGON N5
27: MOVE N5 ZOG N2
28: MOVE N2 ZOG N1
29: MOVE N1 ZOG N12
30: MOVE N12 ZOG N17
31: PICK-UP PEPPERMINT N17 ZOG
32: MOVE N17 ZOG N12
33: MOVE N12 ZOG N11
34: MOVE N11 ZOG N9
35: MOVE N9 ZOG N10
36: DELIVERY ZOG PEPPERMINT BLUE-DRAGON N10
37: DELIVERY ZOG BANDAGE BLUE-DRAGON N10
38: MOVE N10 ZOG N9
39: MOVE N9 ZOG N1
40: MOVE N1 ZOG N2
41: MOVE N2 ZOG N3
42: MOVE N3 ZOG N13
43: MOVE N13 ZOG N14
44: MOVE N14 ZOG N15
45: PICK-UP PLASTER N15 ZOG
46: MOVE N15 ZOG N16
47: PICK-UP BANDAGE N16 ZOG
48: MOVE N16 ZOG N15
49: MOVE N15 ZOG N14
50: MOVE N14 ZOG N4
51: MOVE N4 ZOG N5
52: MOVE N5 ZOG N6
53: DELIVERY ZOG PLASTER ORANGE-DRAGON N6
54: MOVE N6 ZOG N7
55: DELIVERY ZOG STETHOSCOPE GREEN-DRAGON N7
56: MOVE N7 ZOG N6
57: MOVE N6 ZOG N1
58: MOVE N1 ZOG N2
59: MOVE N2 ZOG N3
60: MOVE N3 ZOG N4
61: MOVE N4 ZOG N18
62: MOVE N18 ZOG N19
63: PICK-UP STETHOSCOPE N19 ZOG
64: MOVE N19 ZOG N18
65: MOVE N18 ZOG N4
66: MOVE N4 ZOG N5
67: MOVE N5 ZOG N6
68: MOVE N6 ZOG N7
69: MOVE N7 ZOG N8
70: DELIVERY ZOG STETHOSCOPE YELLOW-DRAGON N8
71: DELIVERY ZOG BANDAGE YELLOW-DRAGON N8
72: MOVE N8 ZOG N7
73: MOVE N7 ZOG N6

```

74: MOVE N6 ZOG N1
75: MOVE N1 ZOG N2
76: MOVE N2 ZOG N3
77: MOVE N3 ZOG N13
78: MOVE N13 ZOG N14
79: MOVE N14 ZOG N15
80: MOVE N15 ZOG N16
81: PICK-UP BANDAGE N16 ZOG
82: MOVE N16 ZOG N15
83: MOVE N15 ZOG N14
84: MOVE N14 ZOG N4
85: MOVE N4 ZOG N5
86: MOVE N5 ZOG N2
87: MOVE N2 ZOG N1
88: MOVE N1 ZOG N12
89: MOVE N12 ZOG N11
90: DELIVERY ZOG BANDAGE PURPLE-DRAGON N11
91: MOVE N11 ZOG N9
92: MOVE N9 ZOG N1
93: MOVE N1 ZOG N12
94: MOVE N12 ZOG GIFT-SHOP
95: BUY-GIFT ZOG GIFT-SHOP

```

```

time spent:    0.00 seconds instantiating 54 easy, 0 hard action templates
              0.00 seconds reachability analysis, yielding 42 facts and 54
actions
              0.00 seconds creating final representation with 42 relevant
facts, 0 relevant fluents
              0.00 seconds computing LNF
              0.00 seconds building connectivity graph
              0.00 seconds searching, evaluating 1977 states, to a max
depth of 0
              0.00 seconds total time

```

=====

```

[barnett]s2634187: ./ff -E -g 1 -h 10 -o domain-1.pddl -f problem-1-
hard.pddl

```

```

ff: parsing domain file
domain 'ZOGS-WORLD' defined
... done.
ff: parsing problem file
problem 'ZOGS-PROBLEM' defined
... done.

```

no metric specified. plan length assumed.

checking for cyclic := effects --- OK.

```

ff: search configuration is  best-first on 1*g(s) + 10*h(s) where
    metric is  plan length

```

```

advancing to distance:  33
                       32
                       31
                       29
                       28

```


27
26
25
24
23
22
21
20
19
18
17
16
15
14
13
12
10
9
8
7
6
5
4
3
2
1
0

ff: found legal plan as follows

```
step    0: MOVE N9 ZOG N1
        1: MOVE N1 ZOG N12
        2: MOVE N12 ZOG N17
        3: PICK-UP PEPPERMINT N17 ZOG
        4: MOVE N17 ZOG N12
        5: MOVE N12 ZOG N11
        6: MOVE N11 ZOG N9
        7: MOVE N9 ZOG N1
        8: MOVE N1 ZOG N2
        9: MOVE N2 ZOG N3
       10: MOVE N3 ZOG N13
       11: MOVE N13 ZOG N14
       12: MOVE N14 ZOG N15
       13: PICK-UP PLASTER N15 ZOG
       14: MOVE N15 ZOG N16
       15: PICK-UP BANDAGE N16 ZOG
       16: MOVE N16 ZOG N15
       17: MOVE N15 ZOG N14
       18: MOVE N14 ZOG N4
       19: MOVE N4 ZOG N18
       20: MOVE N18 ZOG N19
       21: PICK-UP STETHOSCOPE N19 ZOG
       22: MOVE N19 ZOG N18
       23: MOVE N18 ZOG N4
       24: MOVE N4 ZOG N5
       25: DELIVERY ZOG PEPPERMINT PINK-DRAGON N5
       26: MOVE N5 ZOG N2
       27: MOVE N2 ZOG N1
       28: MOVE N1 ZOG N12
       29: MOVE N12 ZOG N17
       30: PICK-UP PEPPERMINT N17 ZOG
```

31: MOVE N17 ZOG N12
32: MOVE N12 ZOG N11
33: MOVE N11 ZOG N9
34: MOVE N9 ZOG N10
35: DELIVERY ZOG PEPPERMINT BLUE-DRAGON N10
36: MOVE N10 ZOG N9
37: MOVE N9 ZOG N1
38: MOVE N1 ZOG N6
39: MOVE N6 ZOG N7
40: DELIVERY ZOG STETHOSCOPE GREEN-DRAGON N7
41: MOVE N7 ZOG N6
42: MOVE N6 ZOG N1
43: MOVE N1 ZOG N2
44: MOVE N2 ZOG N3
45: MOVE N3 ZOG N4
46: MOVE N4 ZOG N18
47: MOVE N18 ZOG N19
48: PICK-UP STETHOSCOPE N19 ZOG
49: MOVE N19 ZOG N18
50: MOVE N18 ZOG N4
51: MOVE N4 ZOG N5
52: DELIVERY ZOG PLASTER PINK-DRAGON N5
53: MOVE N5 ZOG N6
54: MOVE N6 ZOG N7
55: MOVE N7 ZOG N8
56: DELIVERY ZOG STETHOSCOPE YELLOW-DRAGON N8
57: DELIVERY ZOG BANDAGE YELLOW-DRAGON N8
58: MOVE N8 ZOG N7
59: MOVE N7 ZOG N6
60: MOVE N6 ZOG N1
61: MOVE N1 ZOG N2
62: MOVE N2 ZOG N3
63: MOVE N3 ZOG N13
64: MOVE N13 ZOG N14
65: MOVE N14 ZOG N15
66: PICK-UP PLASTER N15 ZOG
67: MOVE N15 ZOG N16
68: PICK-UP BANDAGE N16 ZOG
69: MOVE N16 ZOG N15
70: MOVE N15 ZOG N14
71: MOVE N14 ZOG N4
72: MOVE N4 ZOG N5
73: MOVE N5 ZOG N6
74: DELIVERY ZOG PLASTER ORANGE-DRAGON N6
75: MOVE N6 ZOG N1
76: MOVE N1 ZOG N12
77: MOVE N12 ZOG N11
78: DELIVERY ZOG BANDAGE PURPLE-DRAGON N11
79: MOVE N11 ZOG N9
80: MOVE N9 ZOG N1
81: MOVE N1 ZOG N2
82: MOVE N2 ZOG N3
83: MOVE N3 ZOG N13
84: MOVE N13 ZOG N14
85: MOVE N14 ZOG N15
86: MOVE N15 ZOG N16
87: PICK-UP BANDAGE N16 ZOG
88: MOVE N16 ZOG N15
89: MOVE N15 ZOG N14
90: MOVE N14 ZOG N4
91: MOVE N4 ZOG N5

```

92: MOVE N5 ZOG N2
93: MOVE N2 ZOG N1
94: MOVE N1 ZOG N9
95: MOVE N9 ZOG N10
96: DELIVERY ZOG BANDAGE BLUE-DRAGON N10
97: MOVE N10 ZOG N9
98: MOVE N9 ZOG N1
99: MOVE N1 ZOG N12
100: MOVE N12 ZOG GIFT-SHOP
101: BUY-GIFT ZOG GIFT-SHOP

```

```

time spent:    0.00 seconds instantiating 54 easy, 0 hard action templates
              0.00 seconds reachability analysis, yielding 42 facts and 54
actions
              0.00 seconds creating final representation with 42 relevant
facts, 0 relevant fluents
              0.00 seconds computing LNF
              0.00 seconds building connectivity graph
              0.00 seconds searching, evaluating 1670 states, to a max
depth of 0
              0.00 seconds total time

```

=====

```

[barnett]s2634187: ./ff -E -g 0 -h 1 -o domain-1.pddl -f problem-1-
hard.pddl

```

```

ff: parsing domain file
domain 'ZOGS-WORLD' defined
... done.
ff: parsing problem file
problem 'ZOGS-PROBLEM' defined
... done.

```

no metric specified. plan length assumed.

checking for cyclic := effects --- OK.

```

ff: search configuration is  best-first on 0*g(s) + 1*h(s) where
    metric is  plan length

```

```

advancing to distance:  33
                       32
                       31
                       30
                       29
                       28
                       27
                       26
                       25
                       24
                       23
                       22
                       21
                       20
                       19
                       18
                       17
                       16
                       15

```

14
12
11
10
9
8
7
6
5
4
3
2
1
0

ff: found legal plan as follows

```
step    0: MOVE N9 ZOG N1
        1: MOVE N1 ZOG N2
        2: MOVE N2 ZOG N3
        3: MOVE N3 ZOG N4
        4: MOVE N4 ZOG N18
        5: MOVE N18 ZOG N19
        6: PICK-UP STETHOSCOPE N19 ZOG
        7: MOVE N19 ZOG N18
        8: MOVE N18 ZOG N4
        9: MOVE N4 ZOG N5
       10: MOVE N5 ZOG N2
       11: MOVE N2 ZOG N3
       12: MOVE N3 ZOG N13
       13: MOVE N13 ZOG N14
       14: PICK-UP COUGH-SYRUP N14 ZOG
       15: MOVE N14 ZOG N15
       16: PICK-UP PLASTER N15 ZOG
       17: PICK-UP HEALING-BALM N15 ZOG
       18: MOVE N15 ZOG N16
       19: PICK-UP BANDAGE N16 ZOG
       20: MOVE N16 ZOG N15
       21: MOVE N15 ZOG N14
       22: MOVE N14 ZOG N4
       23: MOVE N4 ZOG N5
       24: MOVE N5 ZOG N2
       25: MOVE N2 ZOG N1
       26: MOVE N1 ZOG N12
       27: MOVE N12 ZOG N17
       28: PICK-UP PEPPERMINT N17 ZOG
       29: MOVE N17 ZOG N12
       30: MOVE N12 ZOG N11
       31: MOVE N11 ZOG N9
       32: MOVE N9 ZOG N10
       33: DELIVERY ZOG PEPPERMINT BLUE-DRAGON N10
       34: MOVE N10 ZOG N9
       35: MOVE N9 ZOG N1
       36: MOVE N1 ZOG N12
       37: MOVE N12 ZOG N17
       38: PICK-UP PEPPERMINT N17 ZOG
       39: MOVE N17 ZOG N12
       40: MOVE N12 ZOG N11
       41: MOVE N11 ZOG N9
       42: MOVE N9 ZOG N1
       43: MOVE N1 ZOG N2
```

44: MOVE N2 ZOG N3
45: MOVE N3 ZOG N4
46: MOVE N4 ZOG N5
47: DELIVERY ZOG PEPPERMINT PINK-DRAGON N5
48: MOVE N5 ZOG N2
49: MOVE N2 ZOG N1
50: MOVE N1 ZOG N12
51: MOVE N12 ZOG N17
52: PICK-UP PEPPERMINT N17 ZOG
53: MOVE N17 ZOG N12
54: MOVE N12 ZOG N11
55: MOVE N11 ZOG N9
56: MOVE N9 ZOG N1
57: MOVE N1 ZOG N2
58: MOVE N2 ZOG N3
59: MOVE N3 ZOG N4
60: MOVE N4 ZOG N5
61: DELIVERY ZOG PLASTER PINK-DRAGON N5
62: MOVE N5 ZOG N2
63: MOVE N2 ZOG N1
64: MOVE N1 ZOG N9
65: MOVE N9 ZOG N10
66: DELIVERY ZOG PEPPERMINT BLUE-DRAGON N10
67: DELIVERY ZOG BANDAGE BLUE-DRAGON N10
68: MOVE N10 ZOG N9
69: MOVE N9 ZOG N1
70: MOVE N1 ZOG N12
71: MOVE N12 ZOG N17
72: PICK-UP PEPPERMINT N17 ZOG
73: MOVE N17 ZOG N12
74: MOVE N12 ZOG N11
75: MOVE N11 ZOG N9
76: MOVE N9 ZOG N1
77: MOVE N1 ZOG N2
78: MOVE N2 ZOG N3
79: MOVE N3 ZOG N13
80: MOVE N13 ZOG N14
81: MOVE N14 ZOG N15
82: PICK-UP PLASTER N15 ZOG
83: MOVE N15 ZOG N16
84: PICK-UP BANDAGE N16 ZOG
85: MOVE N16 ZOG N15
86: MOVE N15 ZOG N14
87: MOVE N14 ZOG N4
88: MOVE N4 ZOG N5
89: MOVE N5 ZOG N6
90: DELIVERY ZOG PLASTER ORANGE-DRAGON N6
91: MOVE N6 ZOG N7
92: DELIVERY ZOG STETHOSCOPE GREEN-DRAGON N7
93: MOVE N7 ZOG N6
94: MOVE N6 ZOG N1
95: MOVE N1 ZOG N2
96: MOVE N2 ZOG N3
97: MOVE N3 ZOG N4
98: MOVE N4 ZOG N18
99: MOVE N18 ZOG N19
100: PICK-UP STETHOSCOPE N19 ZOG
101: MOVE N19 ZOG N18
102: MOVE N18 ZOG N4
103: MOVE N4 ZOG N5
104: MOVE N5 ZOG N6

105: MOVE N6 ZOG N7
106: MOVE N7 ZOG N8
107: DELIVERY ZOG STETHOSCOPE YELLOW-DRAGON N8
108: MOVE N8 ZOG N7
109: MOVE N7 ZOG N6
110: MOVE N6 ZOG N1
111: MOVE N1 ZOG N2
112: MOVE N2 ZOG N3
113: MOVE N3 ZOG N4
114: MOVE N4 ZOG N18
115: MOVE N18 ZOG N19
116: PICK-UP STETHOSCOPE N19 ZOG
117: MOVE N19 ZOG N18
118: MOVE N18 ZOG N4
119: MOVE N4 ZOG N5
120: MOVE N5 ZOG N2
121: MOVE N2 ZOG N3
122: MOVE N3 ZOG N13
123: MOVE N13 ZOG N14
124: MOVE N14 ZOG N15
125: PICK-UP PLASTER N15 ZOG
126: MOVE N15 ZOG N14
127: MOVE N14 ZOG N4
128: MOVE N4 ZOG N5
129: MOVE N5 ZOG N6
130: MOVE N6 ZOG N1
131: MOVE N1 ZOG N9
132: MOVE N9 ZOG N10
133: DELIVERY ZOG PEPPERMINT BLUE-DRAGON N10
134: MOVE N10 ZOG N9
135: MOVE N9 ZOG N1
136: MOVE N1 ZOG N6
137: MOVE N6 ZOG N7
138: MOVE N7 ZOG N8
139: DELIVERY ZOG STETHOSCOPE YELLOW-DRAGON N8
140: DELIVERY ZOG BANDAGE YELLOW-DRAGON N8
141: MOVE N8 ZOG N7
142: MOVE N7 ZOG N6
143: MOVE N6 ZOG N1
144: MOVE N1 ZOG N2
145: MOVE N2 ZOG N3
146: MOVE N3 ZOG N13
147: MOVE N13 ZOG N14
148: MOVE N14 ZOG N15
149: MOVE N15 ZOG N16
150: PICK-UP BANDAGE N16 ZOG
151: MOVE N16 ZOG N15
152: MOVE N15 ZOG N14
153: MOVE N14 ZOG N4
154: MOVE N4 ZOG N5
155: MOVE N5 ZOG N2
156: MOVE N2 ZOG N1
157: MOVE N1 ZOG N12
158: MOVE N12 ZOG N17
159: PICK-UP PEPPERMINT N17 ZOG
160: MOVE N17 ZOG N12
161: MOVE N12 ZOG N11
162: DELIVERY ZOG BANDAGE PURPLE-DRAGON N11
163: MOVE N11 ZOG N9
164: MOVE N9 ZOG N1
165: MOVE N1 ZOG N12

166: MOVE N12 ZOG GIFT-SHOP
167: BUY-GIFT ZOG GIFT-SHOP

time spent: 0.00 seconds instantiating 54 easy, 0 hard action templates
0.00 seconds reachability analysis, yielding 42 facts and 54
actions
0.00 seconds creating final representation with 42 relevant
facts, 0 relevant fluents
0.00 seconds computing LNF
0.00 seconds building connectivity graph
0.00 seconds searching, evaluating 679 states, to a max
depth of 0
0.00 seconds total time

=====

[barnett]s2634187: ./ff -E -g 5 -h 1 -o domain-1.pddl -f problem-1-
hard.pddl

ff: parsing domain file
domain 'ZOGS-WORLD' defined
... done.
ff: parsing problem file
problem 'ZOGS-PROBLEM' defined
... done.

no metric specified. plan length assumed.

checking for cyclic := effects --- OK.

ff: search configuration is best-first on $5*g(s) + 1*h(s)$ where
metric is plan length

advancing to distance: 33
32
31
30
29
28
27
26
25
24
23
22
21
20
19
18
17
16
15
14
13
12
11
10

9
8
7
6
5
4
3
2
1
0

ff: found legal plan as follows

```
step    0: MOVE N9 ZOG N1
        1: MOVE N1 ZOG N2
        2: MOVE N2 ZOG N3
        3: MOVE N3 ZOG N13
        4: MOVE N13 ZOG N14
        5: MOVE N14 ZOG N15
        6: PICK-UP PLASTER N15 ZOG
        7: MOVE N15 ZOG N16
        8: PICK-UP BANDAGE N16 ZOG
        9: MOVE N16 ZOG N15
       10: MOVE N15 ZOG N14
       11: MOVE N14 ZOG N4
       12: MOVE N4 ZOG N18
       13: MOVE N18 ZOG N19
       14: PICK-UP STETHOSCOPE N19 ZOG
       15: MOVE N19 ZOG N18
       16: MOVE N18 ZOG N4
       17: MOVE N4 ZOG N5
       18: DELIVERY ZOG PLASTER PINK-DRAGON N5
       19: MOVE N5 ZOG N6
       20: MOVE N6 ZOG N7
       21: DELIVERY ZOG STETHOSCOPE GREEN-DRAGON N7
       22: MOVE N7 ZOG N6
       23: MOVE N6 ZOG N1
       24: MOVE N1 ZOG N12
       25: MOVE N12 ZOG N17
       26: PICK-UP PEPPERMINT N17 ZOG
       27: MOVE N17 ZOG N12
       28: MOVE N12 ZOG N11
       29: DELIVERY ZOG BANDAGE PURPLE-DRAGON N11
       30: MOVE N11 ZOG N9
       31: MOVE N9 ZOG N1
       32: MOVE N1 ZOG N2
       33: MOVE N2 ZOG N3
       34: MOVE N3 ZOG N13
       35: MOVE N13 ZOG N14
       36: MOVE N14 ZOG N15
       37: PICK-UP PLASTER N15 ZOG
       38: MOVE N15 ZOG N16
       39: PICK-UP BANDAGE N16 ZOG
       40: MOVE N16 ZOG N15
       41: MOVE N15 ZOG N14
       42: MOVE N14 ZOG N4
       43: MOVE N4 ZOG N18
       44: MOVE N18 ZOG N19
       45: PICK-UP STETHOSCOPE N19 ZOG
       46: MOVE N19 ZOG N18
       47: MOVE N18 ZOG N4
```



```

48: MOVE N4 ZOG N5
49: DELIVERY ZOG PEPPERMINT PINK-DRAGON N5
50: MOVE N5 ZOG N6
51: DELIVERY ZOG PLASTER ORANGE-DRAGON N6
52: MOVE N6 ZOG N7
53: MOVE N7 ZOG N8
54: DELIVERY ZOG STETHOSCOPE YELLOW-DRAGON N8
55: DELIVERY ZOG BANDAGE YELLOW-DRAGON N8
56: MOVE N8 ZOG N7
57: MOVE N7 ZOG N6
58: MOVE N6 ZOG N1
59: MOVE N1 ZOG N2
60: MOVE N2 ZOG N3
61: MOVE N3 ZOG N13
62: MOVE N13 ZOG N14
63: MOVE N14 ZOG N15
64: MOVE N15 ZOG N16
65: PICK-UP BANDAGE N16 ZOG
66: MOVE N16 ZOG N15
67: MOVE N15 ZOG N14
68: MOVE N14 ZOG N13
69: MOVE N13 ZOG N3
70: MOVE N3 ZOG N2
71: MOVE N2 ZOG N1
72: MOVE N1 ZOG N12
73: MOVE N12 ZOG N17
74: PICK-UP PEPPERMINT N17 ZOG
75: MOVE N17 ZOG N12
76: MOVE N12 ZOG N11
77: MOVE N11 ZOG N9
78: MOVE N9 ZOG N10
79: DELIVERY ZOG PEPPERMINT BLUE-DRAGON N10
80: DELIVERY ZOG BANDAGE BLUE-DRAGON N10
81: MOVE N10 ZOG N9
82: MOVE N9 ZOG N1
83: MOVE N1 ZOG N12
84: MOVE N12 ZOG GIFT-SHOP
85: BUY-GIFT ZOG GIFT-SHOP

```

```

time spent:    0.00 seconds instantiating 54 easy, 0 hard action templates
              0.00 seconds reachability analysis, yielding 42 facts and 54
actions
              0.00 seconds creating final representation with 42 relevant
facts, 0 relevant fluents
              0.00 seconds computing LNF
              0.00 seconds building connectivity graph
              68.94 seconds searching, evaluating 611375 states, to a max
depth of 0
              68.94 seconds total time

```

=====

```

barnettjs2634187: ./ff -E -g 10 -h 1 -o domain-1.pddl -f problem-1-
hard.pddl

```

```

ff: parsing domain file
domain 'ZOGS-WORLD' defined
... done.
ff: parsing problem file
problem 'ZOGS-PROBLEM' defined

```

... done.

no metric specified. plan length assumed.

checking for cyclic := effects --- OK.

ff: search configuration is best-first on $10*g(s) + 1*h(s)$ where
metric is plan length

advancing to distance: 33
32
31
30
29
28
27
26
25
24
23
22
21
20
19
18
17
16
15
14
13
12
11
10
9
8
7
6
5
4
3
2
1
0

ff: found legal plan as follows

step	0: MOVE N9 ZOG N1
	1: MOVE N1 ZOG N2
	2: MOVE N2 ZOG N3
	3: MOVE N3 ZOG N13
	4: MOVE N13 ZOG N14
	5: MOVE N14 ZOG N15
	6: PICK-UP PLASTER N15 ZOG
	7: MOVE N15 ZOG N16
	8: PICK-UP BANDAGE N16 ZOG
	9: MOVE N16 ZOG N15
	10: MOVE N15 ZOG N14
	11: MOVE N14 ZOG N4
	12: MOVE N4 ZOG N18
	13: MOVE N18 ZOG N19

14: PICK-UP STETHOSCOPE N19 ZOG
15: MOVE N19 ZOG N18
16: MOVE N18 ZOG N4
17: MOVE N4 ZOG N5
18: DELIVERY ZOG PLASTER PINK-DRAGON N5
19: MOVE N5 ZOG N6
20: MOVE N6 ZOG N7
21: DELIVERY ZOG STETHOSCOPE GREEN-DRAGON N7
22: MOVE N7 ZOG N6
23: MOVE N6 ZOG N1
24: MOVE N1 ZOG N12
25: MOVE N12 ZOG N17
26: PICK-UP PEPPERMINT N17 ZOG
27: MOVE N17 ZOG N12
28: MOVE N12 ZOG N11
29: DELIVERY ZOG BANDAGE PURPLE-DRAGON N11
30: MOVE N11 ZOG N9
31: MOVE N9 ZOG N1
32: MOVE N1 ZOG N2
33: MOVE N2 ZOG N3
34: MOVE N3 ZOG N13
35: MOVE N13 ZOG N14
36: MOVE N14 ZOG N15
37: PICK-UP PLASTER N15 ZOG
38: MOVE N15 ZOG N16
39: PICK-UP BANDAGE N16 ZOG
40: MOVE N16 ZOG N15
41: MOVE N15 ZOG N14
42: MOVE N14 ZOG N4
43: MOVE N4 ZOG N18
44: MOVE N18 ZOG N19
45: PICK-UP STETHOSCOPE N19 ZOG
46: MOVE N19 ZOG N18
47: MOVE N18 ZOG N4
48: MOVE N4 ZOG N5
49: DELIVERY ZOG PEPPERMINT PINK-DRAGON N5
50: MOVE N5 ZOG N6
51: DELIVERY ZOG PLASTER ORANGE-DRAGON N6
52: MOVE N6 ZOG N7
53: MOVE N7 ZOG N8
54: DELIVERY ZOG STETHOSCOPE YELLOW-DRAGON N8
55: DELIVERY ZOG BANDAGE YELLOW-DRAGON N8
56: MOVE N8 ZOG N7
57: MOVE N7 ZOG N6
58: MOVE N6 ZOG N1
59: MOVE N1 ZOG N2
60: MOVE N2 ZOG N3
61: MOVE N3 ZOG N13
62: MOVE N13 ZOG N14
63: MOVE N14 ZOG N15
64: MOVE N15 ZOG N16
65: PICK-UP BANDAGE N16 ZOG
66: MOVE N16 ZOG N15
67: MOVE N15 ZOG N14
68: MOVE N14 ZOG N13
69: MOVE N13 ZOG N3
70: MOVE N3 ZOG N2
71: MOVE N2 ZOG N1
72: MOVE N1 ZOG N12
73: MOVE N12 ZOG N17
74: PICK-UP PEPPERMINT N17 ZOG

```

75: MOVE N17 ZOG N12
76: MOVE N12 ZOG N11
77: MOVE N11 ZOG N9
78: MOVE N9 ZOG N10
79: DELIVERY ZOG PEPPERMINT BLUE-DRAGON N10
80: DELIVERY ZOG BANDAGE BLUE-DRAGON N10
81: MOVE N10 ZOG N9
82: MOVE N9 ZOG N1
83: MOVE N1 ZOG N12
84: MOVE N12 ZOG GIFT-SHOP
85: BUY-GIFT ZOG GIFT-SHOP

```

```

time spent:    0.00 seconds instantiating 54 easy, 0 hard action templates
              0.00 seconds reachability analysis, yielding 42 facts and 54
actions
              0.00 seconds creating final representation with 42 relevant
facts, 0 relevant fluents
              0.00 seconds computing LNF
              0.00 seconds building connectivity graph
              67.31 seconds searching, evaluating 620766 states, to a max
depth of 0
              67.31 seconds total time

```

=====

```

[barnett]s2634187: ./ff -E -g 1 -h 0 -o domain-1.pddl -f problem-1-
hard.pddl

```

```

ff: parsing domain file
domain 'ZOGS-WORLD' defined
... done.
ff: parsing problem file
problem 'ZOGS-PROBLEM' defined
... done.

```

no metric specified. plan length assumed.

checking for cyclic := effects --- OK.

```

ff: search configuration is  best-first on 1*g(s) + 0*h(s) where
    metric is  plan length

```

```

advancing to distance:  33
                       32
                       31
                       30
                       29
                       28
                       27
                       26
                       25
                       24
                       23
                       22
                       21
                       20
                       19
                       18

```

17
16
15
14
13
12
11
10
9
8
7
6
5
4
3
2
1
0

ff: found legal plan as follows

```
step    0: MOVE N9 ZOG N1
        1: MOVE N1 ZOG N2
        2: MOVE N2 ZOG N3
        3: MOVE N3 ZOG N13
        4: MOVE N13 ZOG N14
        5: MOVE N14 ZOG N15
        6: PICK-UP PLASTER N15 ZOG
        7: MOVE N15 ZOG N16
        8: PICK-UP BANDAGE N16 ZOG
        9: MOVE N16 ZOG N15
       10: MOVE N15 ZOG N14
       11: MOVE N14 ZOG N4
       12: MOVE N4 ZOG N18
       13: MOVE N18 ZOG N19
       14: PICK-UP STETHOSCOPE N19 ZOG
       15: MOVE N19 ZOG N18
       16: MOVE N18 ZOG N4
       17: MOVE N4 ZOG N5
       18: MOVE N5 ZOG N6
       19: MOVE N6 ZOG N7
       20: DELIVERY ZOG STETHOSCOPE GREEN-DRAGON N7
       21: MOVE N7 ZOG N6
       22: DELIVERY ZOG PLASTER ORANGE-DRAGON N6
       23: MOVE N6 ZOG N1
       24: MOVE N1 ZOG N12
       25: MOVE N12 ZOG N17
       26: PICK-UP PEPPERMINT N17 ZOG
       27: MOVE N17 ZOG N12
       28: MOVE N12 ZOG N11
       29: MOVE N11 ZOG N9
       30: MOVE N9 ZOG N10
       31: DELIVERY ZOG PEPPERMINT BLUE-DRAGON N10
       32: DELIVERY ZOG BANDAGE BLUE-DRAGON N10
       33: MOVE N10 ZOG N9
       34: MOVE N9 ZOG N1
       35: MOVE N1 ZOG N2
       36: MOVE N2 ZOG N3
       37: MOVE N3 ZOG N13
       38: MOVE N13 ZOG N14
       39: MOVE N14 ZOG N15
```

40: PICK-UP PLASTER N15 ZOG
 41: MOVE N15 ZOG N16
 42: PICK-UP BANDAGE N16 ZOG
 43: MOVE N16 ZOG N15
 44: MOVE N15 ZOG N14
 45: MOVE N14 ZOG N4
 46: MOVE N4 ZOG N5
 47: DELIVERY ZOG PLASTER PINK-DRAGON N5
 48: MOVE N5 ZOG N2
 49: MOVE N2 ZOG N1
 50: MOVE N1 ZOG N12
 51: MOVE N12 ZOG N17
 52: PICK-UP PEPPERMINT N17 ZOG
 53: MOVE N17 ZOG N12
 54: MOVE N12 ZOG N11
 55: DELIVERY ZOG BANDAGE PURPLE-DRAGON N11
 56: MOVE N11 ZOG N9
 57: MOVE N9 ZOG N1
 58: MOVE N1 ZOG N2
 59: MOVE N2 ZOG N3
 60: MOVE N3 ZOG N13
 61: MOVE N13 ZOG N14
 62: MOVE N14 ZOG N15
 63: MOVE N15 ZOG N16
 64: PICK-UP BANDAGE N16 ZOG
 65: MOVE N16 ZOG N15
 66: MOVE N15 ZOG N14
 67: MOVE N14 ZOG N4
 68: MOVE N4 ZOG N18
 69: MOVE N18 ZOG N19
 70: PICK-UP STETHOSCOPE N19 ZOG
 71: MOVE N19 ZOG N18
 72: MOVE N18 ZOG N4
 73: MOVE N4 ZOG N5
 74: DELIVERY ZOG PEPPERMINT PINK-DRAGON N5
 75: MOVE N5 ZOG N6
 76: MOVE N6 ZOG N7
 77: MOVE N7 ZOG N8
 78: DELIVERY ZOG STETHOSCOPE YELLOW-DRAGON N8
 79: DELIVERY ZOG BANDAGE YELLOW-DRAGON N8
 80: MOVE N8 ZOG N7
 81: MOVE N7 ZOG N6
 82: MOVE N6 ZOG N1
 83: MOVE N1 ZOG N12
 84: MOVE N12 ZOG GIFT-SHOP
 85: BUY-GIFT ZOG GIFT-SHOP

time spent: 0.00 seconds instantiating 54 easy, 0 hard action templates
 0.00 seconds reachability analysis, yielding 42 facts and 54
 actions
 0.00 seconds creating final representation with 42 relevant
 facts, 0 relevant fluents
 0.00 seconds computing LNF
 0.00 seconds building connectivity graph
 18.93 seconds searching, evaluating 628818 states, to a max
 depth of 0
 18.93 seconds total time